

George Mason University

Smarter Nutrition Through Data

A Machine-Learning Approach to Classifying Food Products by Nutri-Score

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Machine Learning Applications in Science

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2. Introduction

Many consumers struggle to interpret the nutrition facts label on a food product due to lack of time and/or knowledge to comprehensively read the nutrition facts panel of any packaged food. In order to provide assistance regarding interpretation of nutrition facts panels, Nutri-Score was created a system that assigns a letter grade (letter A through letter E) to a food based on its nutrient composition. The purpose of this research project is to determine if a machine learning algorithm can accurately assign a Nutri-Score to a food product using only the nutrient data obtained from the OpenFoodFacts dataset. The project will investigate and evaluate several models to identify the most accurate predictor(s) of the Nutri-Scores.

2.1 Background & Why Automating Nutri-Score Prediction Matters

Nutri-Score is a rating that allows consumers to easily determine the healthiness of a food item by assigning it a grade from A to E based on the overall composition, or profile, of a food item with respect to its sugar, salt, saturated fat, fiber, protein, and energy content. It allows consumers to quickly make decisions about purchasing food without having to look up the nutritional values of each food item.

Providing an automated way to estimate the Nutri-Score is beneficial to consumers because many food items may not have a label on them, and currently calculating the Nutri-Score for a product requires detailed nutrient criteria. Using a Machine Learning system to provide an estimate of Nutri-Scores based on the nutritional information would enable more consumers to be aware of this rating system and have an easier time accessing the Nutri-Score, would result in better and equal comparisons of food items and that would lead to easier usage of Nutri-Scores in consumer apps or even research tools.

2.2 Research Question & Objectives

This study aims to determine whether the Nutrition Score of any given food item can be predicted from a list of its current nutritional values. I have proposed several machine learning algorithms to assist us in creating models to answer this question. In addition to comparing the accuracy and consistency of each method when it came to the predictions they generated for different levels of Nutrition Score, I also looked at what specifically, among all of the nutritional values, contributed to predicting Nutrition Score using each machine learning algorithm so that we could establish their viability and applicability for use in automated Nutrition Score prediction.

3. Data & Preprocessing

3.1 Dataset Overview

I utilized the OpenFoodFacts nutrition dataset for the project and focused on the subset of products that provided both complete nutritional information and a labelled Nutri-Score grade.

The supervised dataset contains information about all of the key nutrients needed including: Energy, Fat, Saturated Fat, Sugar, Carbohydrates, Protein, Fiber, Salt, Vitamins and Minerals as well as categorical attributes such as Food Group and Additive (e.g. Preservatives).

Once the incomplete entries are filtered out or filled with the median or filled with “Unknown”, the final dataset contains over 275,000 products representing numerous nutritional profiles from many different food categories. The distribution between the five classes (A-E) of the Nutri-Score is relatively even and therefore the models are able to learn about each class without the need for resampling of classes.

3.2 Data Cleaning

The dataset required significant preprocessing prior to any analysis. The first step was to assess missingness/completeness of each variable. Any columns that had 100% missingness were dropped right away. Then, relevant columns with greater than 80% missingness, primarily obscure vitamins or micronutrients that would rarely be included in manufacturer summary reports were dropped. This process created a significant reduction in features and retained the most relevant variables pertinent to nutritional prediction.

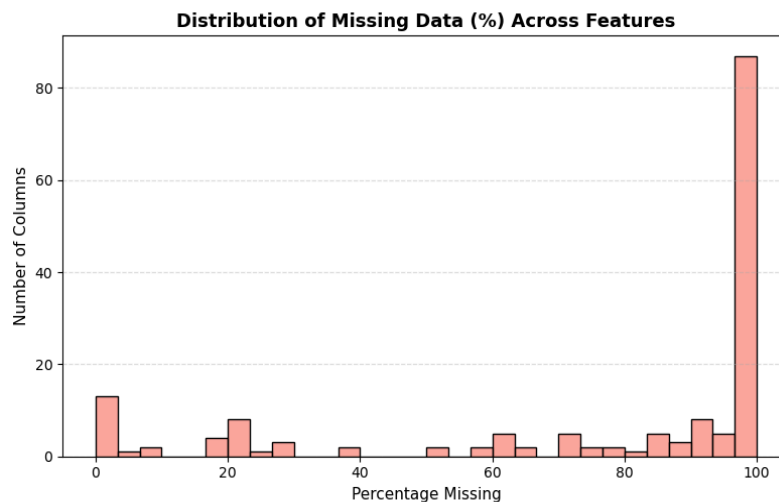


Figure 1: Distribution of missing data from the original dataset

Next, duplicate rows were removed, and irrelevant metadata such as product URLs, packaging text (the product being a legume for example, would be extracted directly from the packaging), and timestamp columns were dropped. Next, I used `pd.to_numeric(errors='coerce')` to convert all nutrient measures into numeric representations. The numeric conversion will yield an 'NaN' for any entries that contain any non-numeric characters. This will facilitate a routine for filling in missing measurements later in the process.

A primary obstacle was overcoming extremes of outlier values in the nutritional variables (stuff like sugars, fats, and energy). I expected that any erroneous values were based in part on manual error, and serving units did not relate correctly for the foods, so threshold values were capped based on what I determined to be reasonable maximum values for nutrition. For example:

- Energy values greater than 20000 kcal/100 grams were capped at 20000.
- Fats, carbs, and proteins were capped at around 100 grams per 100 grams.
- Micronutrients like sodium, Vitamin C, and calcium were capped at reasonable maximum values.

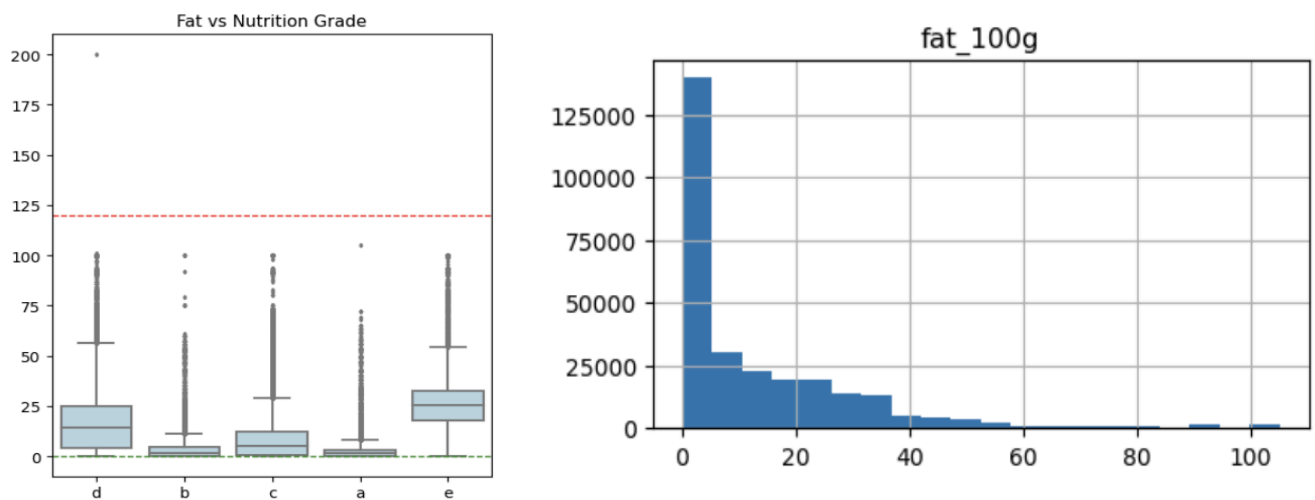


Figure 2: Distribution of the raw `fat_100g` feature before cleaning, showing heavy right-skew and multiple extreme outliers above 100 g per 100 g, which justified median imputation and upper-limit capping.

Figure 3: Boxplot of `fat_100g` across Nutri-Score grades (A–E) after outlier handling, illustrating that foods with higher fat content generally correspond to lower nutrition grades.

This capped maximum value allows the model to learn without being largely affected by incorrect or unrealistic values.

Regarding imputation for missing values, median imputation was used rather than mean imputation. The majority of nutrient variables were highly skewed with very heavy right tails, so the median was a better measure of central tendency without being heavily affected by outliers. Missing values for each numerical feature were replaced with the respective column median. Missing values for categorical variables like `pnns_groups_2` were replaced with the mode.

3.3 Feature Engineering & Label Encoding

In order to improve the dataset's predictive feature set, several new features were created to model important nutrition relationships. Ratio-based variables (e.g., `fat_to_energy`, `protein_to_energy`, `sugar_to_carb`) were computed to represent macronutrient contributions to total energy or carbohydrates, providing a more intuitive summary of the composition of food.

These engineered features provided balance to the nutritional scales and introduced proportional context that could complement the raw nutrient measures. Categorical variables were encoded based on an appropriate context for the intended model to ensure numeric compatibility:

- Frequency Encoding reduced high-cardinality variables (`brands_tags`, `countries_en`) to their relative category frequencies, keeping some information on distribution while reducing a dimension.
- Count Encoding changed the `additives_tags` into `additives_count`, summing the number of additives each product contained.
- One-Hot Encoding changed `pnns_groups_1` into binary indicators of group membership, dropping one base member indicator to prevent redundancy.
- Label Encoding changed the subgroup categories (`pnns_groups_2`) into numeric codes appropriate for modeling.

The processed dataset contained 36 fully numeric features after feature engineering and encoding providing a conducive balance of data structure for continued model training and evaluation.

3.4 Multicollinearity Analysis

To assess that the model predictors were independent and did not contribute additional instability, I conducted an assessment of multicollinearity utilizing a correlation heatmap and Variance Inflation Factor (VIF).

The heatmap expanded the correlations visually to show all of the major nutrient and derived ratios. While the correlation heatmap had a few moderate correlations, the most noteworthy correlation was between energy_100g and fat_100g with $r \approx 0.77$.

This correlation, while relatively high, did not exceed the general linear predictive threshold of 0.8. Thus, the data suggest no predictors led the linear relationships. To verify this visual trend, VIF was tested. All VIF scores were ≤ 6 , most ranging from 2 and 3. This is well within the critical limit of 10.

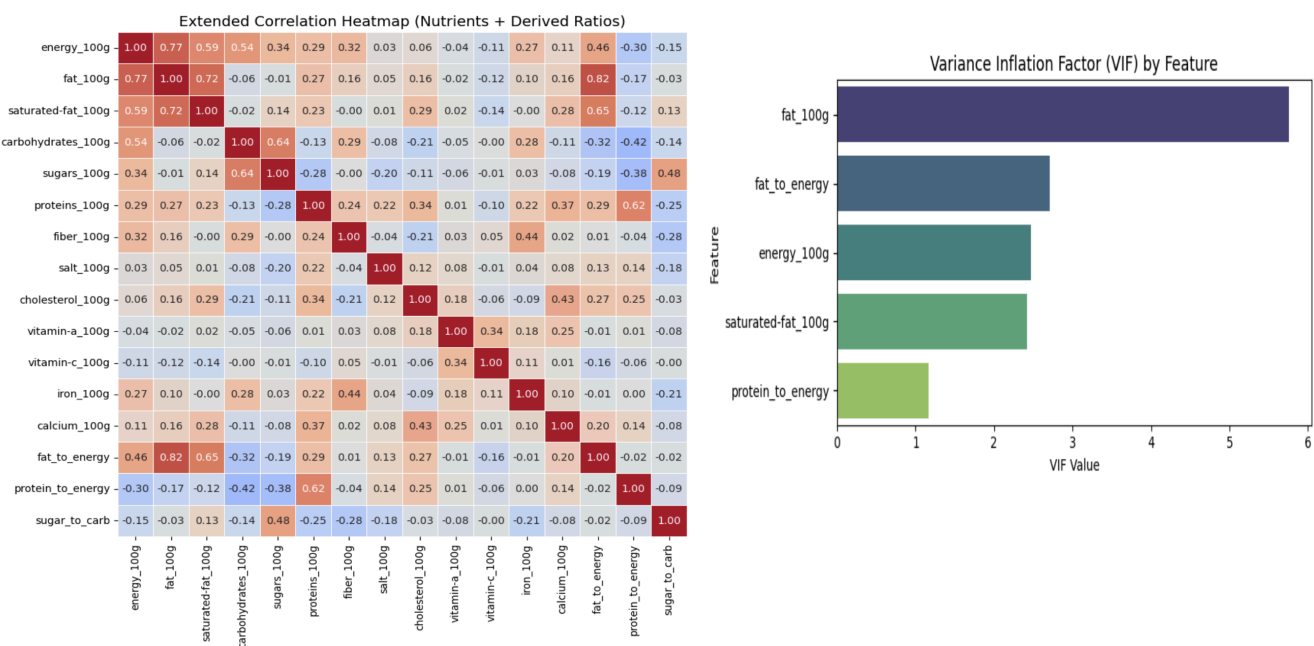


Figure 4: Correlation heatmap to test for multicollinearity

Figure 5: Variance Inflation Factor (VIF) values for key nutrient and ratio features, confirming all multicollinearity levels are within acceptable thresholds ($VIF < 6$)

Overall, there was no evidence suggesting removal or regularization of predictors, all nutrients and ratio features remained in the modeling phase.

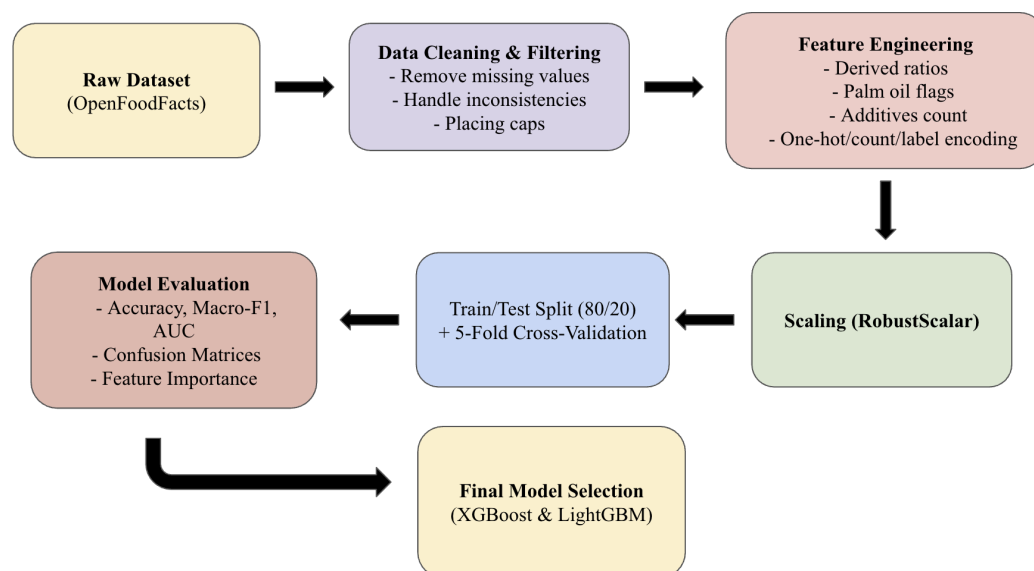
3.5 Scaling & Normalization Strategy

The numeric feature set consists mainly of foods containing nutrients with a large variance in measurements from item to item. I used RobustScaler as a solution for normalizing these measurements to improve the overall stability of the model. Unlike standard normalization or

min-max scaling techniques, we center each numerical feature using an intercept and IQR about each of the values.

By centering the numerical values (meaning the median) and then dividing by IQR, it created a very stable and consistent set of features to work with when training our classifier. Since nutrition data often include extreme outlier values (e.g., high-sugar and fat-containing items), using RobustScaler enables our models to work with the same distribution of features consistently which really enhances the overall performance and reliability of both logistic regression and MLP classifiers.

4. Methodology & Models



4.1 Train/Test Split & Reproducibility

An 80/20 approach was implemented for the stratification of Nutri-Score classes in order to maintain as much balance as possible among class distributions when splitting the data into training and test segments. The random seed had been set at 42 to allow for complete reproducibility of this work by any user. Rather than holding out a validation dataset, I employed cross-validation via five folds on the training data to yield a more precise estimate of the model's ability to generalize, while not affecting the test set until we conducted a final evaluation.

4.2 Models Evaluated (Baseline & Advanced)

As a part of the exploratory phase of our research, we evaluated several models with a variety of learning techniques to identify which models provided greater predictive abilities in predicting

nutrition-related variables. I began our investigation with baseline models so we could compare these to other models for more valid findings. I initially worked with Logistic Regression as it was simple to understand and therefore easier to interpret; however, due to its ability to only model linear relationships between nutrients for the purpose of predicting Nutri-Score, the performance on Nutri-Scores using nutrient values was poor, averaging only 68% accuracy for test data.

I next moved on to tree-based predictive modelling techniques the first being Random Forest. I opted for Random Forest due to its ability to easily create models that model both nonlinear and mixed-feature correlations. It had excellent performance on training data with incredible accuracy, but unfortunately, it had poor predictive power on test data (around ~6% lower). Utilizing hyperparameter tuning (max_depth, min_samples_split, min_samples_leaf, etc.), I was able to reduce the overfitting from a ~6% gap to a ~2% gap. The tuned model achieved 91.8% accuracy and 0.915 macro F1 Score. The tuned model shows improved performance over the untuned, making the model more stable, reliable, and generalizable on newer datasets.

To achieve an even better performance, I evaluated gradient boosting models, specifically XGBoost and LightGBM, which are both highly effective on the type of tabular data found in OpenFoodFacts. As well, both models achieved accuracy rates greater than 97%, macro-F1 scores over 0.97, and had very small amounts of overfitting (around ~2%). A stacking ensemble which included Random Forest, XGBoost, and logistic regression was also evaluated; it performed well, however, the performance wasn't better than two top-performing boosting algorithms.

Overall, for both accuracy and reliability, the top two models were XGBoost and LightGBM and they both had exceptional performance with respect to the accuracy of their predictions, as well as their ability to generalize, and provide balanced performance across the four classes.

4.3 Cross-Validation & Metrics

Stratified Cross Validation by 5-Folds was used to evaluate the training dataset with the goal of confirming the robustness of the different models and not overfitting or depending on only one train/test split. Stratified Cross Validation also enabled evaluation of generalization performance without having to rely on separate validation datasets to evaluate them. The performance of the different models was evaluated against the different accuracy measures, macro F1, macro AUC (Area Under the Curve), ROC curves for multiple classes, and confusion matrices. These

different measures provided insight into the complete performance characteristics of all of the models combined, the relative balance between classes of Nutri-Score categories performed by all models, and how well those models discriminate among five Nutri-Score categories.

5. Results & Model Comparison

5.1 Model Performance Comparison

Every model that was used for comparison, regardless of which one was selected, XGBoost and LightGBM provide the best, most uniform results, achieving a minimum accuracy of 97% and macro F1 score of greater than 0.97 for testing. These were the only two algorithms that outperformed the tuned Random Forest (~91.8% accuracy) and they blew away the base Logistic Regression model that had only ~58% accuracy. You can clearly see from the entire table of results that gradient-boosted decision trees are the winner when it comes to predicting the Nutri Score from nutritional data.

Model	Training Accuracy	Testing Accuracy	Training Macro F1	Testing Macro F1
Logistic Regression	0.5891	0.5829	0.5946	0.5887
Random Forest	0.9383	0.9185	0.9348	0.9152
LightGBM	0.9941	0.9743	0.9940	0.9735
XGBoost	0.9969	0.9731	0.9967	0.9722
Stacking	0.9967	0.9728	0.9965	0.9719

5.2 Confusion Matrices & ROC Curves

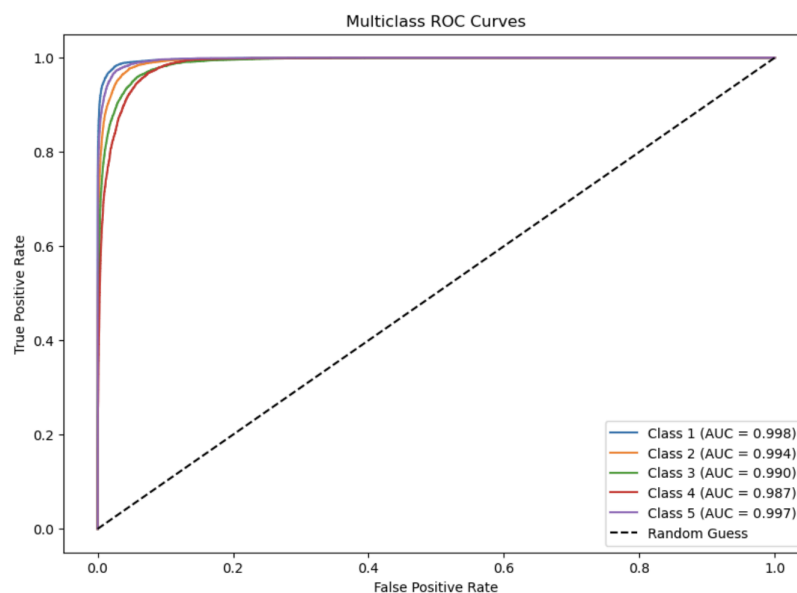
Across all five Nutri-Score classes, XGBoost and LightGBM performed exceptionally well with consistent results. As shown in the confusion matrixes for both models, there are clean diagonals with only a few minor errors with the adjacent classes, we should expect this because the Nutri-Score classifies foods according to their nutritional quality. A tuned Random Forest also had an impressive overall performance; however, it exhibited slightly more confusion between adjacent classes in the middle range (classes 2 and 3) than the boosting models.

For XGBoost, the classification accuracy is highest for the three largest groups (1, 2 and 3) and errors occur mainly between adjacent classes with really no extreme errors.

Class 2 shows the same pattern as XGBoost with LightGBM: near-perfect diagonal classification accuracy, and virtually all errors are found in adjacent classes. There are no extreme errors for this class.

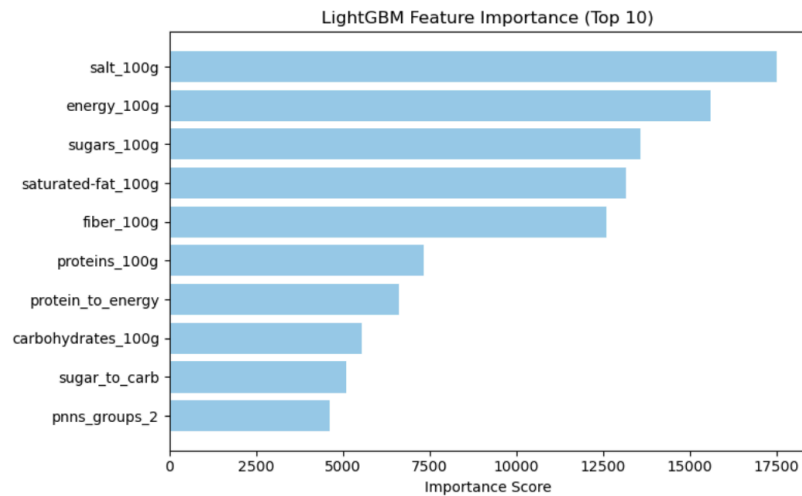
In the end, XGBoost provides the best separation among classes while LightGBM provides the clearest confusion matrix and overall performances of these two models are comparable.

The analysis of the multiclass ROC curve indicated high accuracy, with all five Nutri-Score classes' curves found close to the upper left corner on the ROC plot, so this suggests that there is high true positive and low false positive rates. The AUC values from .987 to .998 showed consistent performance on the model's part for differentiating between the Nutri-Score classes. A macro AUC of .9931 indicates accurate calibration of model output and provides a greater level of divide between classes.



5.3 Feature Importance Summary

Feature Importance of LightGBM model shows that it has been heavily influenced by core nutritional characteristics that directly correlate to the components of Nutri-Score. It places the most weight on Salt, Energy, Sugars, Saturated Fat and Fiber. In addition, Protein and a number of ratios derived from Protein to Energy and Sugar to Carbohydrate were also shown to be helpful features, indicating that there may be more complex relationships between the nutrients than what was captured through the simple nutrient alone.



6.1 Interpretation of Results

Results show that it is possible to construct a model capable of predicting Nutri-Score based on fundamental nutrition features. The confusion matrices demonstrate that nearly all errors could be attributed to the two classes adjacent to one another, thus validating the way these adjacent food items would be consumed daily. Additionally, due to the relatively low occurrence of false-positive predictions, it is evident that both models effectively learned how nutrients interact to affect how food products receive a Nutri-Score designation. Overall, these algorithms correctly identify nutrient types based on the criteria established in the official Nutri-Score calculation, making them suitable tools for developing future models of consumer nutrition.

6.2 Limitations & Future Improvements

The models we have created are built on data from OpenFoodFacts, which can contain missing information or discrepancies in how nutrients are reported. In addition, we used a 1-time train-test set, as opposed to having a validation set that would allow for a more robust approach to the tuning of hyperparameters. Furthermore, there may have been redundancies created through the formation of engineered features.

For the future, we see potential improvements by having a training set (70%) and separate validation (15%) and testing (15%) datasets, performing a more thorough exploration of hyperparameters, and testing the final models against another database of foods to ensure that the final products are as robust as possible. We could also try enhancing the models with more

categorical attributes about foods (for example: type of product, level of processing) so that we can help with borderline cases.

6.3 Final Answer to Research Question

Using data related to nutrition, ML can predict Nutri-Score grade very accurately and consistently by classifying foods into one of five classes (A, B, C, D or E) with proven repeatability. These results were produced using the LightGBM and XGBoost model which also proved their ability to identify key nutrients that impact the Nutri-Score algorithm. The results are a clear demonstration that the generation of Nutri-Score predictions based on a machine-learning model has been achieved via an automated process.

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